

How Do People Actually Prompt? Prompt Smells in 493 Real ChatGPT Requests

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Abstract—People write prompts every day, but we know little about how well they write them outside the lab. We took 493 English user prompts from WildChat, a public log of real ChatGPT conversations, and asked an LLM to label each one with ten *prompt smells*: signs that a request is hard to answer well. Seven in ten prompts (71.4%) had at least one smell. Missing context was by far the most common problem (58.4%), followed by an unstated output format (18.7%) and references such as “this” or “it” that point to nothing in the prompt (16.6%). Smells were most frequent at the two ends of the length scale. Very short prompts were mostly vague, while very long prompts tended to be bloated or to set rules that clash. We release the labeled data and the tool so others can check and extend the results.

Index Terms—prompt engineering, prompt smells, large language models, WildChat, empirical study

I. INTRODUCTION

Most advice on writing prompts comes from guides and from lab studies with set tasks. We know much less about how people write prompts when nobody is watching. WildChat [1] closes part of that gap. It holds about a million real conversations that users had with ChatGPT in exchange for free access.

In code, a *smell* is a sign that something may be wrong, even if the program still runs [2]. We borrow the idea for prompts. A prompt smell is a pattern that makes a request harder for a model to answer well, such as a missing goal, an unclear “it”, or two rules that cannot both hold.

We ask two questions. **RQ1**: How common are prompt smells in real English prompts, and which ones come up most? **RQ2**: Do smell rates change with prompt length?

This short paper reports early results from an ongoing study. Our contributions are (1) a catalog of ten prompt smells with plain definitions, (2) smell rates for 493 real prompts, broken down by length, and (3) an open tool and labeled data set that anyone can rerun.

II. RELATED WORK

White et al. [3] collect prompt patterns that work, and the Prompt Report [4] surveys prompting methods. Both describe what good prompts look like rather than what users really write. Zamfirescu-Pereira et al. [5] watched non-experts write prompts and found they often give up after one failed try and lean on human-style instructions. Our study looks at the same problem at a larger scale, using logs instead of a lab. We use an LLM as the labeler, following the LLM-as-a-judge line of work [6].

III. METHOD

A. Data and sampling

We used one of the six training shards of WildChat (train-00003-of-00006, 88,238 conversations). To avoid taking only the earliest conversations in the file, we kept a conversation when a hash of its id fell in the lowest 1%. This gives a fixed, repeatable sample spread across the whole shard. From it we took every user turn tagged as English that was not empty, 1,003 prompts in all. The labeler works through them in file order. This paper covers the first 493 prompts it labeled. They come from 238 conversations held between 2 and 15 July 2023, and every one of them was sent to GPT-3.5-Turbo. The median prompt is 139 characters long. Labeling of the remaining prompts is still running.

B. Smell catalog

We used ten smells, shown in Table I. They cover missing context (vague prompts, unclear references, no stated output format), structure (one prompt asking for many things, bloat, repetition), contradictions (conflicting constraints), and tone (excess persona, loaded framing, wrong register).

C. LLM labeling

Each prompt was sent on its own to gpt-oss-120b [7] through the Groq API at temperature 0. The instructions list the ten smells with a short definition, mark the prompt with fixed start and end markers, and say whether the prompt is the first turn or a follow-up, so that a short follow-up such as “make it shorter” is not marked vague just because it leans on earlier turns. The model had to return JSON with a smell name and a one-sentence reason for each smell found. We checked every answer against a schema and asked again, up to three attempts in all, when it did not fit. Close spellings of a smell name were mapped to the catalog name. Prompts longer than 12,000 characters were cut, and the model was told so. Prompts that still failed were logged and retried in a later pass; the 493 prompts here are those with a valid label.

IV. RESULTS

A. RQ1: How common are smells?

Of the 493 prompts, 352 (71.4%) had at least one smell and 141 (28.6%) had none. 166 prompts had one smell, 133 had two, and 53 had three or more. Across all prompts the mean was 1.25 smells, or 1.76 among prompts with any smell.

TABLE I
PROMPTS WITH EACH SMELL ($n = 493$). A PROMPT CAN HAVE SEVERAL.

Smell	Prompts	%
Vague / Missing Context	288	58.4
Format Ambiguity	92	18.7
Ambiguous References	82	16.6
Prompt Bloat / Convolved Prompt	45	9.1
Overloaded Prompt	31	6.3
Conflicting Constraints	31	6.3
Unnecessary Repetition	28	5.7
Bias / Loaded Framing	11	2.2
Irrelevant / Excessive Persona	10	2.0
Formality / Audience Mismatch	0	0.0
Any smell	352	71.4

TABLE II
SMELL RATE BY PROMPT LENGTH IN CHARACTERS.

Length	Prompts	With smell (%)	Mean smells
< 50	91	81.3	1.16
50–199	195	68.2	1.12
200–999	121	61.2	1.11
$\geq 1,000$	86	82.6	1.85

Table I shows how often each smell appeared. Missing context led by a wide margin: it was found in 58.4% of prompts, more than the next three smells put together. Format ambiguity (18.7%) and ambiguous references (16.6%) came next. Tone problems were rare. Loaded framing and excess persona each showed up in about 2% of prompts, and the model never flagged a formality mismatch. The labeler was allowed to invent new smell names but did not do so.

Smells tended to come in pairs. Of the 82 prompts with an ambiguous reference, 78 were also marked vague, and 62 prompts had both a vague goal and an unclear format. Of the 31 overloaded prompts, 22 were also bloated.

B. RQ2: Does length matter?

Table II splits prompts by length. The pattern is U-shaped. Prompts under 50 characters had a high smell rate (81.3%), the rate fell to 61.2% for prompts of 200–999 characters, and it rose again to 82.6% for prompts of 1,000 characters or more. The two ends fail in different ways. Among short prompts, 80.2% were vague and 27.5% had an ambiguous reference. Among long prompts the mix was wider: 36.0% were bloated, 33.7% vague, 30.2% had conflicting constraints and 26.7% left the format unclear. Long prompts also carried the most smells each (1.85 on average).

V. DISCUSSION

The main lesson is simple: most real prompts do not say enough. People seem to write to ChatGPT the way they would talk to a colleague who already knows the task, leaving out the goal, the audience or the shape of the answer. The pairing of ambiguous references with vague prompts fits this view. A user writes “fix this” and assumes the model can see what “this” is.

Length is not a fix in itself. Writing more helps up to a point, but very long prompts bring their own problems. They bury the request, repeat it, or set rules that pull against each other. Mid-length prompts, a few sentences with a clear goal, had the lowest smell rate.

For tool builders, this suggests that a light check before sending, such as asking “what format do you want?” or pointing out an “it” with nothing to refer to, could catch many problems. For people who teach prompting, the gains are likely to come from basics, such as stating the goal and the expected output, rather than from advanced tricks.

VI. THREATS TO VALIDITY

Labels. The labels come from one LLM and were not checked by people. The model may be too strict or too lenient, and we did not measure agreement with human raters. The counts should be read as estimates. **Sample.** We used one shard, a 1% sample and English only, and this paper covers only the first 493 of 1,003 sampled prompts. These come from two weeks in July 2023 and were all sent to GPT-3.5-Turbo, so we cannot compare models yet. WildChat language tags are sometimes wrong; we saw a few Chinese prompts tagged as English. WildChat users chose to share their chats for free access, so they may not represent all ChatGPT users. **Context.** Each prompt was judged alone, with only a first-turn or follow-up flag. A prompt that looks vague alone may be clear in its chat.

VII. CONCLUSION

In 493 real ChatGPT prompts, seven in ten showed at least one prompt smell, and missing context was the most common by far. Short prompts tend to be vague, and very long ones tend to be tangled. Next we will finish labeling all 1,003 prompts, compare models, have people label a subset to check the LLM labels, and test whether fixing a smell leads to better answers.

DATA AND CODE AVAILABILITY

The labeled data (`wildchat_493_smells.json`), the summary statistics and the analysis tool are archived on Zenodo at <https://doi.org/10.5281/zenodo.XXXXXXX>. The prompts come from WildChat and are shared under its ODC-BY license.

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